PROJECT

DESIGN AND DEVELOPMENT OF AN AI BASED QUADCOPTER PROTOTYPE FOR ANIMAL OBSERVATION AND WILDLIFE MONITORING

Supervising teacher: Associate Professor. LÊ MỸ HÀ, Ph.D

Student: Trần An Phú

TP. HỒ CHÍ MINH, JULY 2026

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TABLE OF CONTENTS

Chapter 1 Overview and research scope

Chapter 2 Theoretical foundations

Chapter 3 System design and construction

Chapter 4 Wildlife detection and coordinate workflow

Chapter 5 Experiments and evaluation

Chapter 6 Conclusion and future development

References

LIST OF FIGURES

Figure 1.1 Quadcopter concept
Figure 2 UAV history and early autonomous aircraft
Figure 3 Modern aircraft
Figure 4 Quadcopter types
Figure 5 Four-rotor motion
Figure 6 IMU overview
Figure 7 INS system
Figure 8 Gyroscope
Figure 9 GPS system
Figure 10 ArduPilot
Figure 11 MAVLink overview
Figure 12 MAVLink message
Figure 13 Mission Planner
Figure 14 EKF block diagram
Figure 15 PID controller
Figure 16 Object detection example
Figures 17–18 Camera coordinate geometry
Figure 3.1 Functional system architecture
Figures 19–20 UAV frame and prototype
Figure 3.2 Functional connections
Figures 4.1–4.3 Dataset and detection workflow
Figure 5.1 Validation confusion matrix
Figure 5.2 F1-confidence curve
Figures 5.3–5.4 Precision curves
Figures 5.5–5.6 Predictions and monitoring interface

LIST OF ABBREVIATIONS

Abbreviations	Content
UAV	Unmanned Aerial Vehicle
GPS	Global Positioning System
AI	Artificial intelligence
CW	Clockwise
CCW	Counterclockwise
DOF	Degrees of Freedom
PID	Proportional Integral Derivative
IMU	Inertial Measurement Unit
GCS	Ground Control Station
EKF	Extended Kalman Filter

CHAPTER 1: OVERVIEW

This project explores how a camera-equipped quadcopter and computer vision can support animal observation. Tran An Phu's interest in horseback riding, including his observations of changes in horses' gait, helped motivate the project. He built a physical prototype and collected video and field data on horses at a riding school and wildlife-related data, including elephants and gaur, in Ma Da Forest, Dong Nai.

The physical build, field-data collection, detector development and proposed telemetry workflow are distinct parts of the project. The validation figures retained in this report use the dataset labels Crocodylidae and Mew; they do not establish detector performance on the horses, elephants or gaur observed during field work. Animal geolocation and a live onboard pipeline require separate evaluation.

1.1 Problem statement

Wildlife monitoring is important for conservation, ecological research, and forest management. However, traditional methods such as direct observation, field surveys, and camera traps have many limitations. They require significant time and manpower, are difficult to apply in large or remote areas, and may expose field workers to safety risks.

In addition, wild animals often move unpredictably and are difficult to detect in forests, mountains, wetlands, or other natural environments. Fixed camera traps can only monitor limited locations and cannot actively search wider areas.

The project therefore develops a camera-equipped quadcopter prototype and explores a workflow for attaching candidate image detections to UAV telemetry for ground-station review. Autonomous survey operation, reliable animal geolocation and real-time conservation deployment remain development goals.

1.1.1 Conservation monitoring context

Biodiversity monitoring depends on repeated observations that can be compared across places and time. Protected-area managers need records of where target animals were observed, when observations occurred, and how reliable each identification is. The IUCN framework for biodiversity monitoring emphasizes selecting indicators and methods that fit the management question rather than collecting observations without a defined purpose [34].

Ground surveys and camera traps remain essential, but their coverage is shaped by access, placement and staff time. An aerial camera can complement these methods by surveying a planned route and recording candidate animal detections. It does not, by itself, establish population size, species identity beyond the trained labels, or the absence of animals from unobserved habitat.

Table 1.1. Monitoring needs and the proposed system response

Monitoring need	Current constraint	Prototype contribution
Spatial coverage	Ground routes may be slow or inaccessible	Planned aerial image capture
Repeatability	Ad hoc observations are hard to compare	Timestamped route and imagery
Candidate identification	Manual review requires staff time	Two-label object detection for review
Image location	A box gives pixels, not ground coordinates	Bounding-box center plus telemetry
Reliability	False positives and missed targets occur	Validation curves and confusion matrix
Animal welfare	Aircraft can disturb animals	Site protocol and disturbance monitoring
Operational evidence	Validation images differ from field surveys	Separate field and geolocation tests

The table summarizes design requirements for this prototype; it does not report field performance.

The research question is therefore whether a camera-equipped quadcopter and a trained object detector can identify the project's two labeled targets in image frames and produce records that a ground operator can review. Validation of the detector and validation of geographic coordinates are separate tasks.

Wildlife monitoring in Viet Nam

Viet Nam's National Biodiversity Strategy to 2030, with a vision to 2050, establishes conservation and biodiversity management as national priorities [35]. Monitoring tools can support those priorities when deployed under the direction of site managers and combined with existing ecological survey methods.

In forests, wetlands and other conservation areas, an aerial observation workflow may help teams inspect selected open areas or inaccessible locations. Vegetation cover, lighting, flight altitude and animal movement can limit detection. Flights should also minimize disturbance to wildlife and comply with site permissions; the present report does not claim a field survey in a protected area.

Table 1.2. Scope of evidence presented in this report

Evidence available	What it supports
Annotated dataset screenshots	Bounding-box labeling workflow
Confusion matrix and curves	Two-label image validation metrics
Validation image batch	Examples of predicted bounding boxes
Prototype interface screenshot	Concept for operator review
No ground-truth coordinate series	Geolocation accuracy remains unmeasured

The distinction between model validation and operational field validation is retained throughout the report.

The document contains a prototype UAV design, sample interface screens and validation outputs for two dataset labels. It does not include a documented independent field survey, geolocation ground truth, flight endurance measurement or ecological population estimate. Those evaluations are required before the system can be used as a dependable conservation survey tool.

1.2 Related Work

Existing monitoring methods include direct observation, fixed camera traps and aerial surveys. UAVs can add mobile image coverage and timestamped imagery, while computer vision can flag possible targets for human review. The IUCN monitoring framework [34] supports matching a survey method to a specific indicator, and drone use must account for possible animal disturbance [36]. This project examines detection and data reporting at prototype scale.

1.2.1. Purpose

Build a camera-equipped quadcopter prototype, collect animal-observation data and develop an image-detection workflow. Evaluate the available two-label detector separately from field-data collection and proposed location-linked monitoring.

1.2.2. Objectives

The objectives of the project are set as follows:

- Describe the quadcopter configuration and flight control needed for stable image capture along a planned route.
- Prepare annotated images, train a YOLO26 object detector, and evaluate it on the available validation split.
- Extract class labels, confidence scores and bounding-box centers from detections and associate them with time-stamped UAV telemetry.
- Display candidate detections and approximate location information at the ground station for operator review.

1.3 Topic limitations

- The proposed waypoint route assumes a clear flight path; obstacle avoidance and safe operation near wildlife have not been demonstrated.
- The reported detector is limited to the two dataset labels, Crocodylidae and Mew; the exact biological meaning and provenance of “Mew” must be confirmed against the dataset label file.
- The report does not provide ground-truth measurements of animal coordinates, detection performance in a protected-area survey, or population estimates.

1.4 Research methods

This project combines theoretical and experimental research methodologies.

Theoretical research: Review quadcopter flight control, aerial imaging, object detection, camera geometry and the data fields needed to link a detection with UAV telemetry. Define the difference between image localization and verified geographic location.

Experimental research: Use the annotated image dataset to train and validate the YOLO26 detector. Review the confusion matrix, confidence curves, precision-recall curve and sample predictions. Inspect the prototype interface and describe the coordinate-estimation workflow. A separate flight and geolocation test with surveyed ground-truth targets remains future work.

CHAPTER 2: THEORETICAL FOUNDATIONS

This chapter presents the fundamental knowledge necessary for the research and implementation of the project. The content includes an overview of quadcopter systems, an analysis of the structure and operating mechanism of the aircraft. It also introduces control algorithms, image processing techniques applied in the model, the operating principles of the sensors used, and communication standards between the microcontroller and wireless connection methods.

2.1 Overview of drones

2.1.1 Concept of Unmanned Aircraft

A quadcopter is a type of UAV that uses four symmetrically arranged engines to generate stable lift. Unlike traditional aircraft, UAVs operate as intelligent mobile robots, which can be remotely controlled via a ground station or operated autonomously thanks to an integrated computer system with multi-directional sensors and a pre-programmed flight plan based on GPS[5]. This mechanism allows UAVs/quadcopters to operate accurately without manual intervention, expanding the range of applications from monitoring to automated delivery.

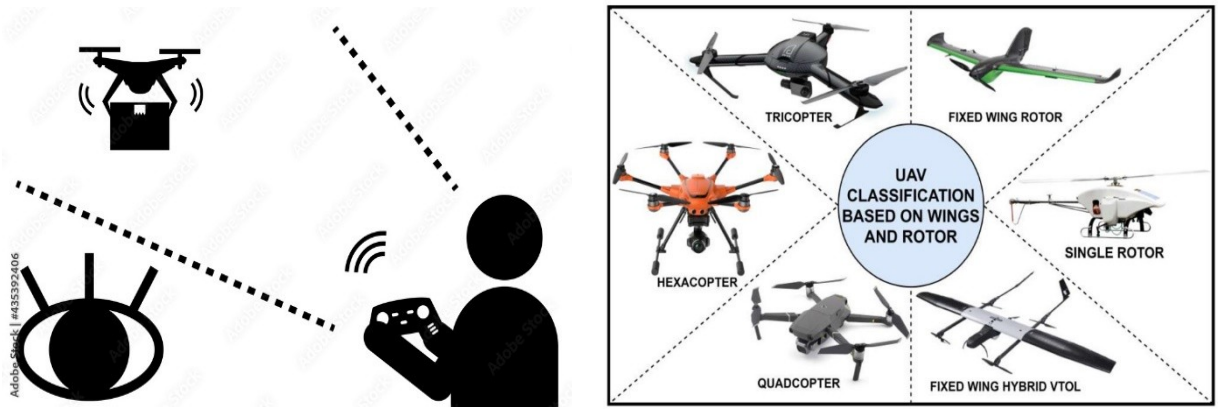


Figure 2.1 Concept of a Quadcopter

2.1.2 History of Origin and Development

Preliminary period (15th century - early 20th century) In the late 15th century, the Italian genius Leonardo da Vinci sketched the idea of the "Aerial Screw" - a spiral-structured flying device considered the earliest precursor to modern helicopters and drones. This drawing demonstrates his timeless vision of the possibility of building aircraft. It was not until the 19th century that the first practical applications appeared when the Austrian army used unmanned airships in military operations[6].

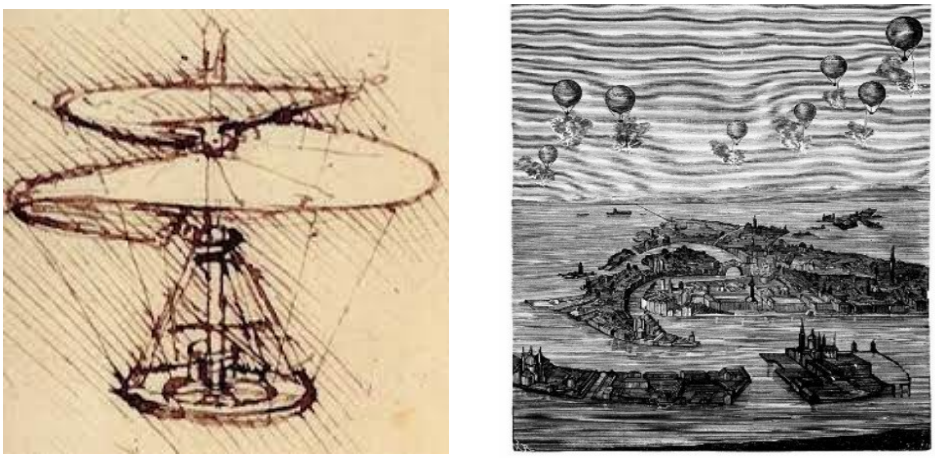


Figure2. History of the development of airplanes.

The period of technological development (20th century) marked an important step forward when the US developed the Kettering Bug - the first automated aircraft used in World War I. The period 1950-1970 saw the birth of modern military UAVs such as the Ryan Firebee, mainly serving the purpose of reconnaissance and surveillance. The appearance of GPS technology in the 1980s opened a new era for civilian drones[7].



The modern drone era (2010-present): The 2010s marked a revolution in civilian drones as unmanned aerial vehicles (UAVs) became widespread thanks to miniaturized microprocessor technology and affordable prices. Brands like DJI and Parrot brought drones to the masses. Notably, 2013 saw a turning point when Amazon announced Project Prime Air, paving the way for drone delivery applications.



Figure 6. Modern aircraft

Recent civilian UAV applications include environmental mapping and conservation observation. For wildlife surveys, route planning and image quality must be balanced against animal disturbance, vegetation cover and the limited endurance of a multirotor aircraft. The detector in this project is evaluated on its validation images; operational survey performance remains to be measured.

2.1.3 Analysis and advantages/disadvantages of the configuration



Figure 8. Some types of quadcopters currently available.

With their diverse technical specifications, operating methods, and applications, UAVs can be classified as follows:

Table 2-1: Analysis of UAV configurations

UAV type	Advantage	Disadvantages
	- Long flight time (energy saving) - High speed, wide operating range	- A runway is needed. - High cost, complex operation. - Sensitive to weather conditions

	- Vertical take-off/landing - Better load capacity than multirotor	- Low stability - Maintenance is expensive. - Requires advanced driving skills.
	- Easy to use, inexpensive - Terrain flexibility - Stable during flight	- Short flight time (<30 minutes) - Low load capacity (<5kg) - Slow speed

Multicopters (multi-rotor aircraft) are currently the top choice for surveillance and object tracking applications due to their flexible operation. In fact, this type of UAV has many different versions, including:

- Tricopter: A 3-motor design with a simple control mechanism.

- Hexacopter: 6 propellers provide a balance between performance and flight stability.

- Octocopter: An 8-motor system that enhances payload capacity and safety.

With many differences in structure and control distribution, a performance comparison between them is necessary.

Table 2-2: Performance Comparison of Multicopter Types

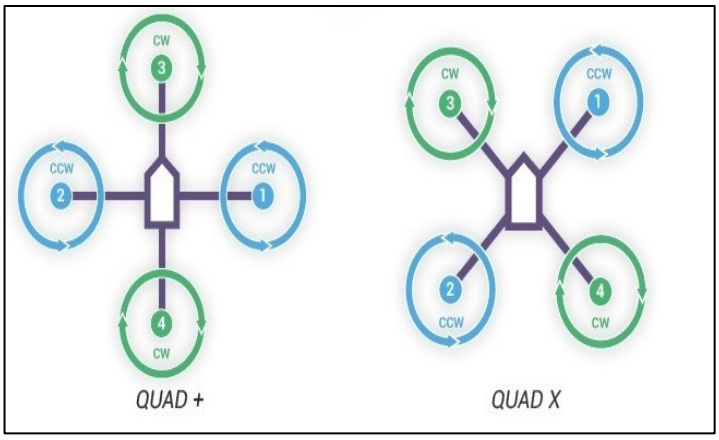
STT	Element	Tricopter	Quadcopter	Hexacopter	Octocopter
1	Number of fan blades	3	4	6	8
2	Stabilization	Adjust	Good	Good	Very good
3	Load	Light	Medium	High	Very high
4	Lifting capacity	Short	Medium	High	Very high
5	Wind resistance	Short	Medium	High	Very high
6	Stability	Short	High	Very high	Very high
7	Complexity	Short	Short	High	Very high

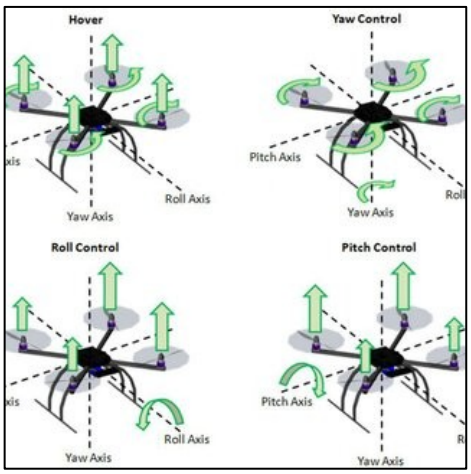
8	Size	Small	Small	Medium	Big
9	Energy consumption	Short	Medium	Medium	High
10	Cost	Short	Short	Medium	High

Quadcopters (four-rotor aircraft) are highly regarded for their superior technical and economic advantages. Structurally, the mechanical design of quadcopters is relatively simple compared to other multi-rotor UAVs. Their compact size allows them to operate easily in confined spaces. In particular, the quadcopter control system is relatively easy to learn, suitable for beginners. Low production costs are also an important factor to consider. In fact, these advantages make quadcopters an optimal solution for student research projects, meeting both technical requirements and limited financial resources.

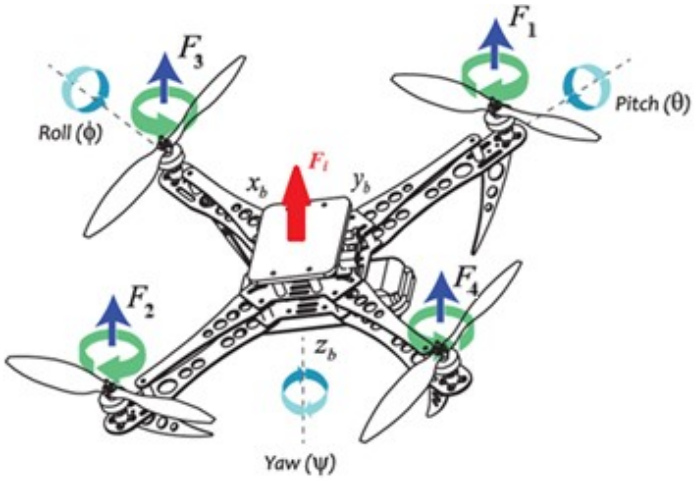
2.1.4 Quadcopter Flight Control Theory

Quadcopters have a cross-shaped, H-shaped, or X-shaped frame and operate based on a system of four brushless motors, each equipped with a propeller to generate lift upon receiving signals from the microcontroller. The key feature lies in the opposing rotational arrangement of the propeller pairs: one pair rotates clockwise (CW) and the other rotates counterclockwise (CCW), which cancels out torque and maintains angular momentum balance. Motion control is achieved by adjusting the differential rotational speed between the propeller pairs - changing the speed of the left/right pair controls the roll angle, adjusting the front/rear pair changes the pitch angle, while the yaw angle is controlled by creating a speed difference between the forward and reverse motors in each pair. This system allows the quadcopter to perform complex flight maneuvers with high precision[9][10][11].



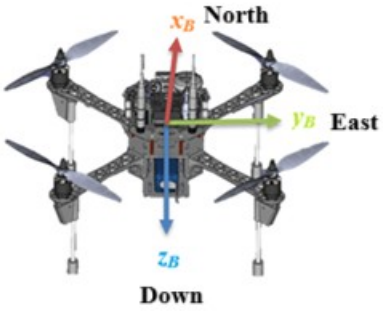


The quadcopter model has a structure consisting of four propellers that generate corresponding angular velocity, momentum, and force, as shown in Figure 11. Two pairs of propellers are attached to brushless motors, with propellers 1 and 2 rotating counterclockwise (CCW), and propellers 3 and 4 rotating clockwise (CW). This arrangement helps balance torque and stabilize the quadcopter's movement in space.



Quadcopters operate on the principle of balancing lift and torque. Four motors generate total lift (F) to overcome gravity (F > mg), while simultaneously canceling out all torque thanks to counter-rotating propellers (CW/CCW). When the motor speeds are balanced, the device flies steadily without rotating its axis

2.1.6 Coordinate System



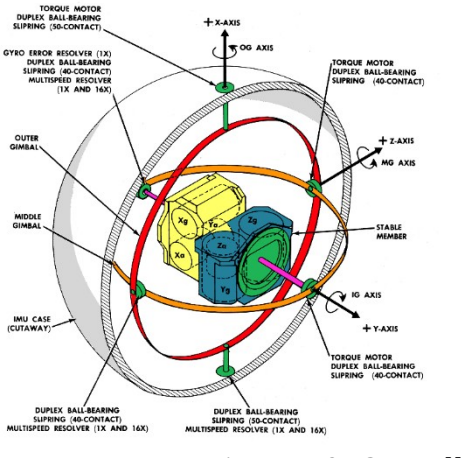
The quadcopter model is 6 DOF (6 degrees of freedom) based on Newton-Euler's laws of motion with twelve state variables forming a system of equations. The figure below illustrates the coordinate system following the NED (North-East-Down) setup rule.

To describe the convention for Quadcopters within the Body frame, it is necessary to establish a mathematical model for converting between: Earth coordinate frame (E), body coordinate frame (B) and vice versa.

2.1.7 Measurement and Navigation Module of the Unmanned Aircraft

2.1.7.1 IMU Sensor

An inertial measurement unit (IMU) is a device that integrates several key sensors to track motion in space. The main components of an IMU include three core elements: an accelerometer to measure changes in speed, a gyroscope to determine the direction of rotation, and a magnetic field sensor to assist with spatial orientation [19] [20]. The data collected from these sensors are processed by the system to accurately calculate the current position based on the parameters of speed of movement and the corresponding time interval.

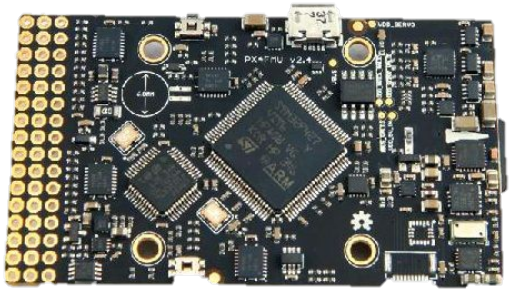


2.1.7.2 INS system

An inertial navigation system (INS) consists of two main components: the IMU (Inertial Measurement Unit) and the navigation computer. The IMU is responsible for collecting data from the accelerometer and gyroscope sensors, while the navigation computer processes coordinate system conversion, calculates gravitational acceleration, applies noise filtering algorithms, and integrates data. By using dead reckoning, the INS allows the vehicle's position and direction of travel to be determined independently of external signals [18].

The main drawback of traditional INS systems is the cumulative measurement error over time, leading to a discrepancy between calculated and actual values. To overcome this, the research team proposes integrating anti-interference modules (such as

open-source control circuits) combined with multiple sensors (accelerometer, magnetic field sensor, GPS) to calibrate the IMU before each startup at different geographical locations. Simultaneously, the dynamic shield design minimizes static pressure interference caused by airflow from the fan, ensuring stability for the measurement

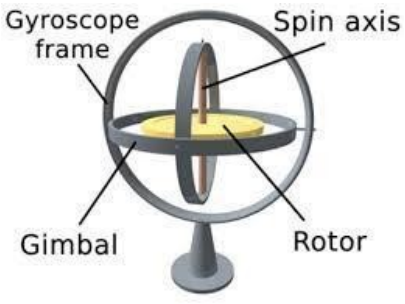


2.1.7.3 Accelerometer

An accelerometer is a sensor used to determine the change in velocity of an object over time. By measuring acceleration, this sensor clearly distinguishes between two components: dynamic acceleration (arising from the actual motion of the object) and static acceleration (caused by field forces such as gravity or friction). In inertial measurement systems, the unit of measurement is usually standardized to g ($g \approx 9.8\ m/s^2$) instead of using m/s^2 directly, simplifying calculations in positioning algorithms.

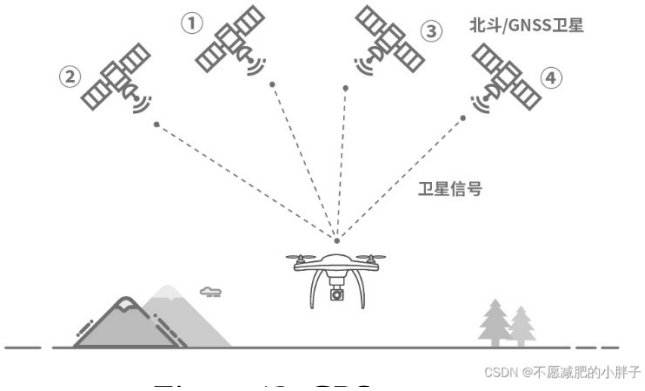
2.1.7.4 Gyroscope

Gyroscopes operate based on the law of conservation of angular momentum to determine the rotational speed of an object. This device records the change in angle along three coordinate axes, thereby providing important data for the control system to stabilize the flight direction of the aircraft under all operating conditions [18].



2.1.7.5 GPS System

The Global Positioning System (GPS) determines aircraft coordinates through signals from a network of artificial satellites orbiting the Earth. By measuring the signal transmission time from at least three satellites (for 2D positioning) or four satellites (for



3D positioning), the receiver can calculate the precise position based on triangulation. In this study, GPS acts as the central sensor for the aircraft positioning system [18].

2.1.8 Flight Control

Ardupilot is a dedicated open-source platform for implementing UAV control algorithms via the Pixhawk board. The system comprises two main modules: a Flight Controller (FC) that integrates PID algorithms to process data from inertial measurement units (IMUs), pressure sensors, and GPS; and a Ground Control System (GCS) that provides a real-time monitoring interface via the MAVLink protocol. The FC performs sensor fusion, combining multi-source data to calculate tilt/declination angles, while the GCS allows for waypoint flight path programming and displays telemetry data on a digital map.

2.1.9 MAVLink Protocol

MAVLink (Micro Air Vehicle Link) is an open-source communication protocol specifically designed for UAV systems, acting as an intermediary in bidirectional data exchange between quadcopters and ground control stations (GCS).

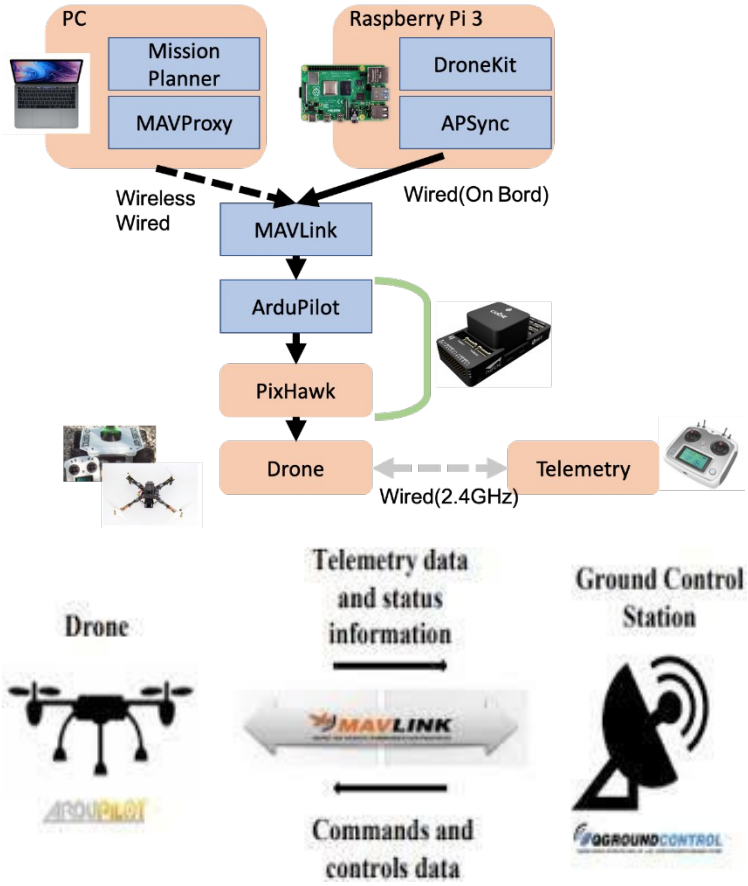


Figure 15. Overview of the MAVLink protocol

MAVLink has a structure consisting of 3 parts: CRC (Cyclic Redundancy Check), Payload, and Header.

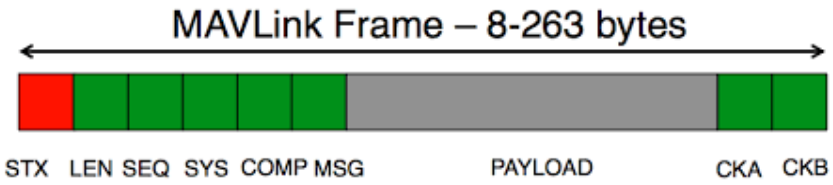


Figure 16. MAVLink message structure

Table 2-3: Description of MAVLink protocol messages

Header			
1	STX	Start of Text	The character marking the beginning of the 0xFE packet.
2	ONLY	Payload Length	Information about the number of bytes of data contained in the packet.
3	SEQ	Packet	The packet order starts from 0 and increases

		Sequence	progressively.
4	SYS	System ID	ID of the packet-sending system
5	COMP	Component ID	ID of the device sending the packet
6	SMG	Message ID	The type of message and data sent in a packet.
Payload			
7		Mission Data	UAV mission: flight path, location, speed
8		Status Flags	Status flags are used to mark the state of the system.
9		Time Stamp	Synchronize and record the time of sending/receiving packets.
10		Fields	The data field contains specific values.
CRC			
11	WHAT		Verify the integrity of the data in the payload.
12	CKB		

2.1.10. DroneKit Library

DroneKit is an open-source application development platform for UAV systems, providing a set of programming libraries (APIs) that allow interaction with the drone through communication protocols. This tool focuses on building intelligent control solutions, enabling integration with onboard computers to perform complex tasks such as programming automatic flight paths, processing real-time sensor data, and synchronizing with ground equipment.

Table 2-4: Architecture of DroneKit

STT	Ingredient	Function	Describe
1	Connection	Managing the connection between the drone and the computer.	Protocol: MAVLink + Drone API Function: Establish a two-way communication channel between drone and PC.
2	Vehicle State	Drone status management	Monitoring: GPS (longitude/latitude), 3-axis speed, tilt angle Update: Real-time
3	Vehicle Control	Control management	Method: Command-based Mode: Manual/Semi-automatic/Automatic

4	Telemetry	Monitor status information	Battery, link and flight-status monitoring; update frequency and latency were not measured.
5	Mission Planning	Flight planning and missions	Waypoint mission planning; an obstacle-avoidance implementation is not documented.

2.1.11 Mission Planner Data Collection Software

Mission Planner is an open-source platform that supports the management and control of drone operations through an intuitive interface. This software allows users to design flight paths, monitor technical parameters (such as altitude, speed, and position) in real time, and analyze data after mission completion. Additionally, it provides direct control over critical operations such as takeoff/landing and flexible flight mode switching, meeting diverse requirements from planning to executing flights with precision.



In this project, the Mission Planner software was deployed on a Windows laptop to support the collection and analysis of flight data. The software's intuitive interface displayed detailed information about the selected operational area – the stadium at Ho Chi Minh City University of Technology and Education – along with technical parameters related to the drone's flight path and range. The software's visual representations not only helped pinpoint the exact location but also ensured accuracy in planning and monitoring the mission in real time.

2.1.12 EKF

The Extended Kalman Filter (EKF) plays a central role in integrating multi-sensor data (IMU, GPS, BARO) to create an accurate dynamic model for the drone, overcoming the cumulative errors of each individual sensor. In particular, the EKF maintains stable altitude estimates even when the GPS signal is interrupted by combining atmospheric

pressure from BARO and acceleration from the IMU. The resulting estimated altitude (ALT) becomes the basis for the PID controller to adjust engine thrust, ensuring the drone maintains target altitude throughout its flight – a crucial factor for sensitive operations such as delivery or precise landing.

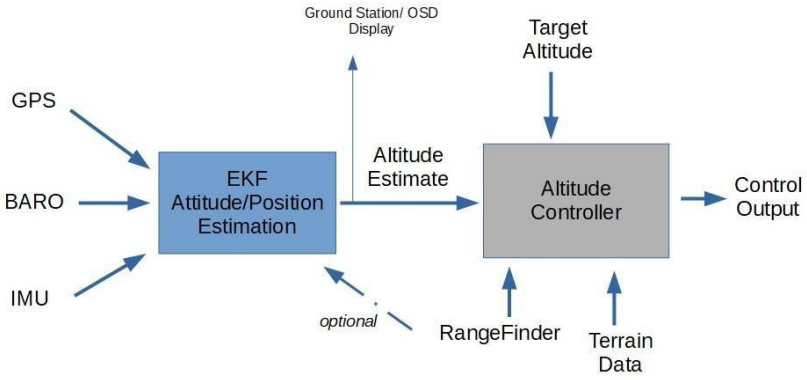


Figure 18. Block diagram of the EKF filter

2.2 PID controller theory

PID (Proportional Integral Derivative) controllers are a widely used closed-loop feedback control solution in industry due to their flexible adjustment capabilities. This mechanism works by calculating the error between the measured value from the system and the preset target value, thereby optimizing the operation process. With advantages in reliability and high performance, PID has become the top choice for systems requiring accuracy and stability [21] [22] [23].

The structure of a PID controller is based on three main components: proportional (P), integral (I), and derivative (D). The proportional component directly responds to the current error, minimizing instantaneous deviation. The integral component handles errors accumulated over time, eliminating residual error. Meanwhile, the derivative component predicts the trend of error change, thereby limiting oscillations and increasing system stability. The harmonious combination of these three elements allows PID to balance fast response and accuracy, meeting diverse requirements in automatic control.

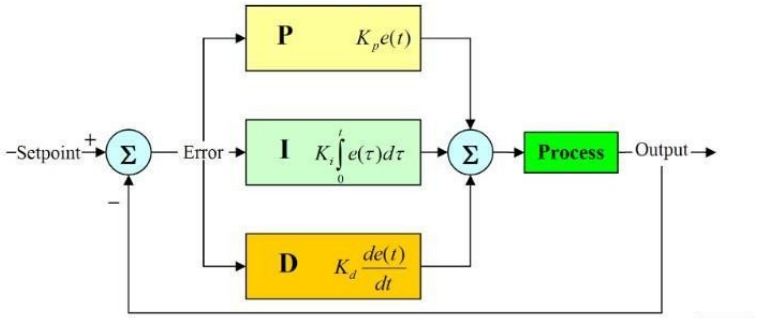


Figure 19. PID control block diagram

PID controllers do not necessarily require the simultaneous use of all three components (proportional, integral, and derivative). Depending on the system characteristics, users can flexibly choose only one or two control stages by setting the coefficient of the unused stage to zero. For example, a PI controller (combining proportional and integral) is suitable for systems needing to eliminate residual errors, while a PD controller (proportional and derivative) helps reduce rapid oscillations. However, adjusting the parameters requires careful consideration: the derivative component (D) is easily affected by measurement noise, and reducing the integral value (I) too low can cause the system to fail to achieve the target value. Therefore, optimizing the P, I, and D coefficients is crucial to ensuring stable and accurate performance for the control system.

2.3 Image processing applicantions

2.3.1 YOLO26 Theory

2.3.1.1. Definition of YOLO26

YOLO26 is an object detection model in the Ultralytics YOLO family [37]. It is used here to predict bounding boxes and labels for the two dataset classes. Its validation outputs are appropriate for measuring performance on labeled images; they do not alone establish performance on unseen habitats.

For each processed image frame, the model returns a target label, confidence score and bounding box. These outputs provide a candidate observation and an image position. Geographic estimation requires synchronized UAV telemetry and calibrated camera geometry.

The project uses the detection output as one stage of a proposed UAV monitoring pipeline. Actual inference speed on the Raspberry Pi and reliability of live onboard operation were not measured in the supplied results.

2.3.1.2. Operating principle

The YOLO26-based detection process begins when the quadcopter camera captures an aerial image of the monitoring area. The image is resized and normalized before being passed

into the YOLO26 model. During inference, the model analyzes the image in a single forward pass and predicts possible wildlife targets.

For each detected animal, YOLO26 returns a bounding box, class label, and confidence score. The bounding box is represented by four image coordinates, usually x1, y1, x2, and y2. From these values, the center point of the detected animal can be calculated as follows:

$$u = (x1 + x2) / 2, \quad v = (y1 + y2) / 2$$

The image center (u, v) is a pixel coordinate. Converting it to a ground coordinate requires camera intrinsics, camera orientation, UAV position, altitude, a suitable ground-plane assumption and error validation. Without those inputs, the UAV GPS position is only a rough reference.

The detector therefore supports image-level identification and target localization. Linking a detected target to a trustworthy geographic coordinate remains a separate calibration and field-test task.

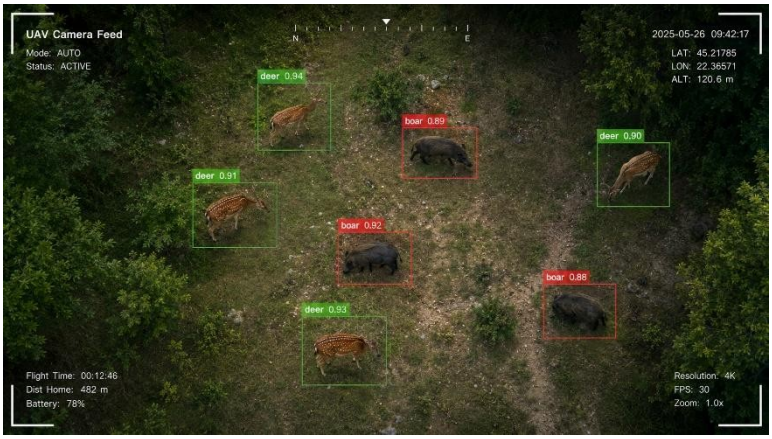


Figure 21. Operating principle

2.3.2. Computer Vision

2.3.2.1. Camera Calibration

The camera's position and orientation relative to the world coordinate system are called its external parameters. Meanwhile, how the camera maps perspective points in real space onto the image plane (such as focal length and optical center) are called its internal parameters. The process of determining both these external and internal parameters is camera calibration. Camera calibration establishes a mathematical relationship between the 3D coordinates of objects in the real world and their 2D coordinates in the image, thereby enabling applications such as distance measurement,

3D reconstruction, or robot navigation to operate accurately. For example, focal length is a crucial internal parameter that determines the camera's field of view, while position and rotation determine how the camera observes objects in space.

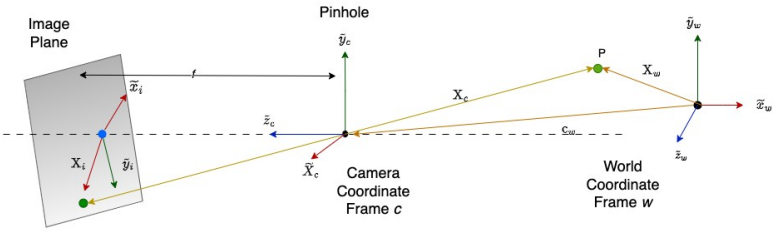


Figure 22. Real 3D and 2D coordinate systems in images

2.3.2.2. Homogenous Coordinates

The process involves mapping a point from a 3D world coordinate system to a 3D camera coordinate system through coordinate transformation. This point is then projected onto a 2D image coordinate system using perspective projection. This process depends on the camera's internal parameters (such as focal length) to determine the mapping scale and external parameters (camera position and orientation) to link the world coordinate system to the camera.

We know that:

Converting images from millimeters to pixels:

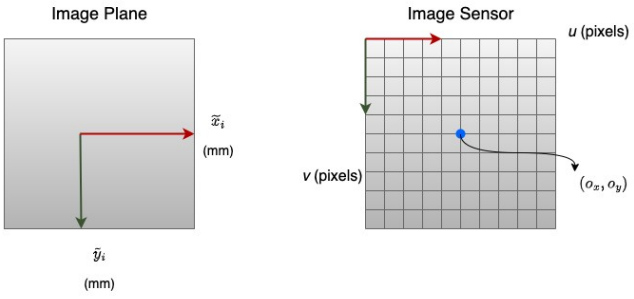


Figure 23. Optical center

In the image coordinate system, if m_x and m_{and}These are the pixel densities (pixels/mm²) in each direction, respectively.*x* and *and*Then the pixel coordinates are determined as follows: The following formulas show the relationship between 3D spatial coordinates and 2D pixel coordinates in a camera system:

In camera systems, the top-left corner of the image sensor is typically chosen as the origin of the coordinate system. Pixel point (*ox*,*oy*) This point is called the Principal Point, where the optical axis passes through the sensor. This location plays a crucial role in determining image coordinates and calculations related to camera calibration:

We have:

While () = (, the internal parameters include: (*fx*,*fy*) is the focal length in pixels in two directions *x* and *y*, determined by the formula (*fx*,*fy*)=(*mxf*,*myf*), in there *mx* and *my* These are the pixel densities in the two directions, respectively and *f* is the physical focal length of the camera.

Parameter set (*fx*,*fy*,*ox*,*oy*) This represents the internal geometry of the camera, including the focal length and keypoint position (*ox*,*oy*), where the optical axis intersects the image plane. These parameters play an important role in converting coordinates from 3D space to 2D image coordinates[8], as well as in camera calibration processes and computer vision applications.

CHAPTER 3: SYSTEM DESIGN AND CONSTRUCTION

This chapter presents an overview of the system, the system design process including the drone model and data collection and monitoring. The model includes mechanical and electrical components. After defining the model requirements, the team searched for suitable equipment. From the selected equipment, the team designed the model using Solidworks software. Next, the team designed the electrical components and wiring diagrams for the quadcopter prototype. After design, the team proceeded with the construction of the quadcopter prototype.

3.1 System Overview

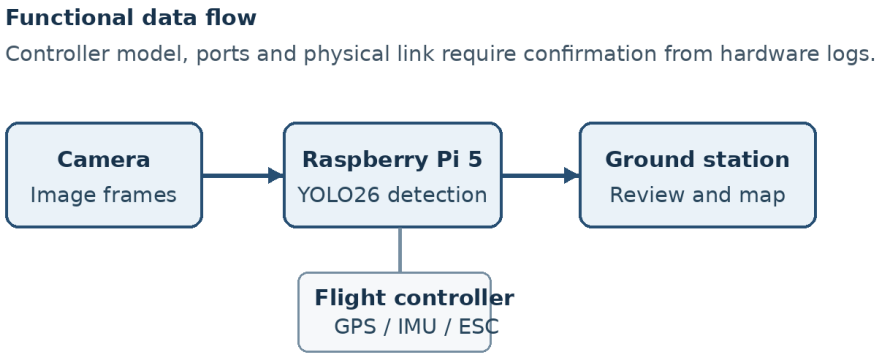


Figure 3.1. Functional architecture of the proposed monitoring system.

The prototype architecture includes a camera, Raspberry Pi 5, flight controller, GPS and a ground monitoring computer. The camera supplies frames to the Raspberry Pi for detection; the flight controller supplies telemetry. The available hardware list names a Goku H743 controller, while older diagrams in the source draft used Pixhawk. The exact installed controller and communication link must be verified from the physical build.

- Raspberry Pi 5: Onboard computer intended to process camera frames and package detection results with telemetry.

- Logitech camera: Imaging source for the detector; the physical connection and model specifications should be verified against the installed unit.

- Flight controller: The equipment table identifies a Goku H743. It stabilizes the vehicle and provides telemetry to the companion computer, subject to the actual

firmware and wiring.

- ESCs: Receive motor commands from the flight controller and regulate the four brushless motors.

- Brushless motors: Four brushless motors provide the flight motion for the quadcopter model, allowing the aircraft to increase or decrease altitude and move in various directions...

3.2 Designing the model and selecting aircraft components

The quadcopter model, based on the DJI-F650 drone, was designed using SolidWorks 3D modeling software.

3.2.1. Quadcopter Aircraft Frame

3.2.1.1. Frame body

The X-shaped airframe is used to connect the components of the Quadcopter [43]. This is an important part in determining the control method and is a criterion for selecting suitable equipment such as motors, propellers, batteries, etc. The frame also directly affects the aircraft's load capacity.



Figure 25. F650 frame

The specific parameters are presented in the table below.

Table 3-1. Aircraft frame dimensions

STT	Ingredient	Specifications (mm)

1	X-shaped size	363x363
2	Size +	495x495
3	Top base	107.5x107.5
4	Bottom base	180x110
5	Base height	40
6	The distance between the two engines	455

Overview of the Quadcopter System:

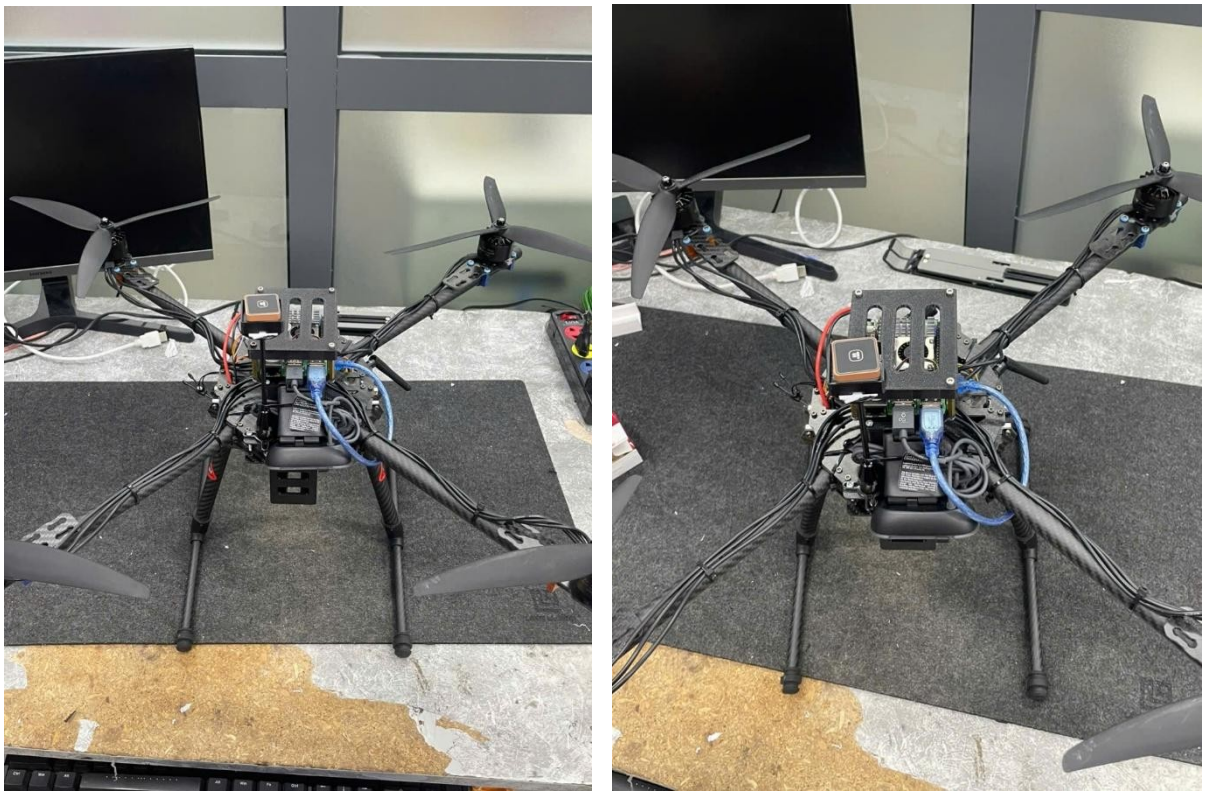
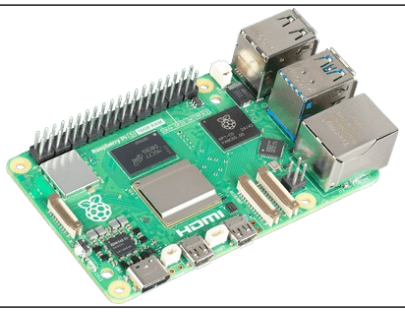


Figure 26. UAV overview

3.2.2. Equipment Components

Bảng 3- 2. Thiết bị bay

STT	Ingredient	Image	Features
1	Raspberry Pi 5 Model B		-Quad core Cortex-A72 64-bit - 3 RAM options: 2, 4, 8GB LPDDR4-2400 SDRAM - 2 USB 3.0 ports and 2 USB 2.0 ports -40 GPIO pins - DC power supply 5V – 3A
2	Goky H743		-The ARM Cortex M4 32-bit MPU6000 core is the primary accelerometer and gyroscope. - 5 ports: UART, I2C, SPI USB, CAN - ADC output: 3.3V and 6.6V -7V power supply
3	BrotherHobby Tornado T5 3115 Pro – 900kv Brushless Motor		- Speed 1100KV - Uses 2-4S batteries -Current 12-24A -Length: 36.5mm -Weight: 72g
4	Tekko 32 F4 Metal 4-in-1 65a Speed Controller		- Input: 3-4S Lipo -BEC output: 5V, 5A -Frequency -Maximum speed: 300,000 RPM Output current: 40A -Weight: 39g
5	Module GPS BE – 880 M10 with Compass		-Power supply: 3.6-5.5V -Accuracy: 1.5m -Baud rate: 4800-921600bps Output frequency: 10Hz

6	Camera Logitech 1080		- Fixed focal length lens -8 megapixel camera - Supports 1080p30, 720p60, and 640x480p90 video resolutions. -Weight: 3g -Rotation angle 62.2º - Connect UART port
7	Pin Lipo		-Capacity: 5500 mAh -Voltage: 14.8V -Dimensions: 13.9 x 4.4 x 3.6cm -Weight: 445g
8	XL4015E 5A DC-DC Buck Converter		Input voltage: 4V-38V Output voltage: 1.25V-36V -Current: 5A

3.3 Wiring diagram

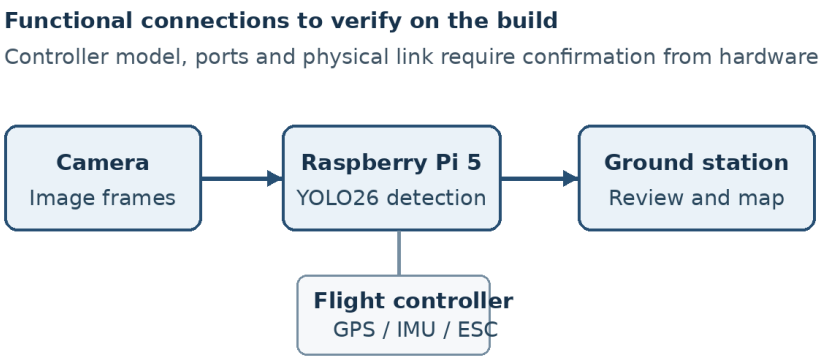


Figure 3.2. Functional connections requiring hardware verification.

The conceptual wiring links the camera to the companion computer, the flight controller to the ESCs and navigation sensors, and the battery through appropriate power regulation. The equipment table names a Goku H743, but the original wiring description named Pixhawk and conflicting camera interfaces. Figure 3.2 is therefore a functional diagram; the physical port map must be confirmed before reproducing the build.

CHAPTER 4: YOLO26-BASED WILDLIFE DETECTION MODEL AND COORDINATE TRANSMISSION

This chapter presents the development of an AI-based wildlife detection model using YOLO26 and a real image dataset prepared on the Roboflow platform. The system is designed for UAV-based monitoring, where animals are detected from aerial camera images, their bounding boxes are extracted, and the corresponding location information is transmitted to the ground control station for observation and recording.

This chapter describes object detection for the two annotated dataset labels and the data fields needed to associate a detection with UAV telemetry. Image-level validation is presented separately from unmeasured field geolocation accuracy.

4.1 YOLO26 Wildlife Detection Model

YOLO26 is used as the main object detection model in this project because it is suitable for real-time image recognition tasks. In the proposed UAV system, each camera frame is processed by the trained YOLO26 model to identify animal targets within the image. For each detected object, the model returns a rectangular bounding box, a class label, and a confidence score.

This detection format is appropriate for UAV applications because the bounding box provides both the approximate position and size of the animal in the image. These values can then be used for target localization, coordinate estimation, and data transmission to the ground control station.

Output component	Description	Use in the system
Class label	Predicted animal category from the trained dataset	Identifies the detected wildlife target
Confidence score	Probability score of the detection result	Filters weak or uncertain detections
Bounding box	Pixel coordinates [x1, y1, x2, y2] around the animal	Used to calculate target center and image error
Frame metadata	UAV GPS, altitude, timestamp, and flight status	Used to attach location information to the detection result

Table 4.1. Main YOLO26 detection outputs used in the UAV wildlife monitoring system.

4.2 Dataset Preparation Using Roboflow

The Roboflow screenshots document a bounding-box annotation workflow. Field-data collection at the riding school and Ma Da Forest is part of the student's project record, but the

supplied validation outputs retain the labels Crocodylidae and Mew. These outputs are treated as a separate detector experiment. The report does not include a manifest linking those figures to the horse or wildlife field footage, or the image counts and train/validation/test split.

Each image was annotated manually by drawing a bounding box around the visible animal target. This step is important because YOLO26 learns the position and category of each object from the annotated bounding boxes. Accurate annotation directly affects the ability of the trained model to detect animals during UAV operation.

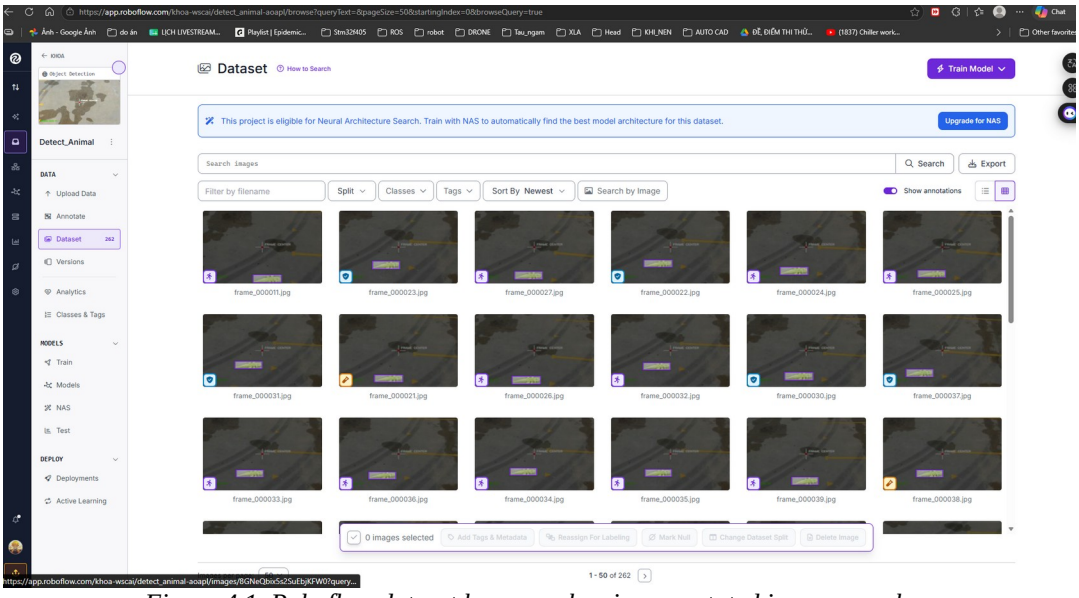


Figure 4.1. Roboflow dataset browser showing annotated image samples.

The project view confirms a bounding-box annotation workflow. The screenshots alone do not establish that the frames were collected during wildlife flights or that the validation split is independent of the training source.

Dataset item	Description
Platform	Roboflow
Task type	Object Detection
Model used	YOLO26
Annotation format	Bounding box annotation
Image source	Annotated image frames; capture site not documented
Main purpose	Two-label object detection on the reported validation images

Table 4.2. Dataset configuration for the YOLO26 wildlife detection model.

4.2.1 Annotation and Labeling Process

During annotation, each animal visible in the image is enclosed by a bounding box. The bounding box must cover the full body of the animal as closely as possible while avoiding unnecessary background area. This helps the model learn the visual characteristics of the target more accurately.

The annotated data are then used to train the YOLO26 model. During training, the model compares its predicted bounding boxes with the ground-truth annotations and updates its parameters to improve detection accuracy.



Figure 4.2. Example of bounding-box annotations in the dataset.

4.2.2 Data Preprocessing and Dataset Versioning

Image resizing and augmentation are possible preprocessing steps in Roboflow, but the exact settings for this training run are not recorded in the supplied report. They should be listed with the dataset version and split when the experiment is reproduced.

Future dataset versions should record image provenance, class definitions, split strategy and preprocessing settings so that results can be reproduced and leakage between similar frames can be checked.

4.3 YOLO26 Training Procedure

The YOLO26 model was trained using the dataset exported from Roboflow. The dataset contains images and corresponding label files, where each label file stores the class index and normalized bounding box coordinates of the animal target. During training, YOLO26 learns to predict the object class and bounding box position directly from the image.

The supplied figures document validation curves and predictions. The exact model variant, training command, epochs, input size, split sizes and hardware should be added from the training log before this is presented as a reproducible experiment.

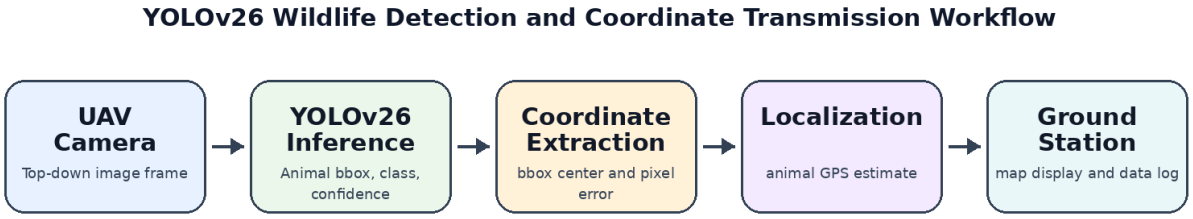


Figure 4.3. Workflow of YOLO26 wildlife detection and coordinate transmission.

Stage	Function
Data export	Export annotated dataset from Roboflow in YOLO-compatible format
Training	Train YOLO26 to detect animal targets from image frames
Validation	Evaluate model performance on unseen validation images
Inference	Run trained model on UAV camera feed
Coordinate extraction	Calculate bounding box center and attach UAV position data
Transmission	Send detected animal information to the ground control station

Table 4.3. Main stages of the YOLO26 training and deployment process.

4.3.1 Evaluation Metrics

Precision, recall, F1 and AP@0.5 describe performance on labeled validation images. The supplied figures do not show mAP@0.5:0.95, so this stricter metric is defined below but is not reported as a result.

Metric	Meaning in wildlife detection
Precision	Measures how many predicted animal detections are correct
Recall	Measures how many real animal targets are successfully detected
F1-score	Balances Precision and Recall into one score
mAP@0.5	Measures average detection accuracy at IoU threshold 0.5
mAP@0.5:0.95	Measures detection performance across stricter IoU thresholds

Table 4.4. Evaluation metrics for the YOLO26 wildlife detection model.

4.4 Bounding Box and Target Coordinate Extraction

The YOLO26 detection result is represented by a rectangular bounding box surrounding the detected animal. The bounding box is defined by four pixel coordinates:

bbox = [x1, y1, x2, y2]

where (x1, y1) represents the top-left corner of the bounding box and (x2, y2) represents the bottom-right corner. These coordinates are measured in the image coordinate system, where the origin is located at the top-left corner of the frame.

4.4.1 Calculating the Bounding Box Center

The center of the detected animal is calculated by averaging the bounding box coordinates. This point represents the approximate position of the animal in the image and is used for target

localization and UAV control logic.

$$xc = (x1 + x2) / 2 \qquad yc = (y1 + y2) / 2$$

where xc and yc are the horizontal and vertical coordinates of the bounding box center.

4.4.2 Pixel Error Relative to the Frame Center

To determine the position of the animal relative to the UAV camera center, the system calculates the pixel error between the bounding box center and the frame center. If the image width is W and the image height is H, the frame center is located at (W/2, H/2).

$$ex = xc - W/2 \qquad ey = yc - H/2$$

The values ex and ey indicate whether the detected animal is located to the left, right, above, or below the center of the camera frame. These values can be used for target centering or for estimating the relative position of the animal on the ground.

4.5 Animal Geographic Coordinate Estimation

A candidate detection can be time-aligned with UAV latitude, longitude, altitude, heading and camera information. Those telemetry fields are inputs to coordinate estimation; the supplied results do not include a measured target-position error.

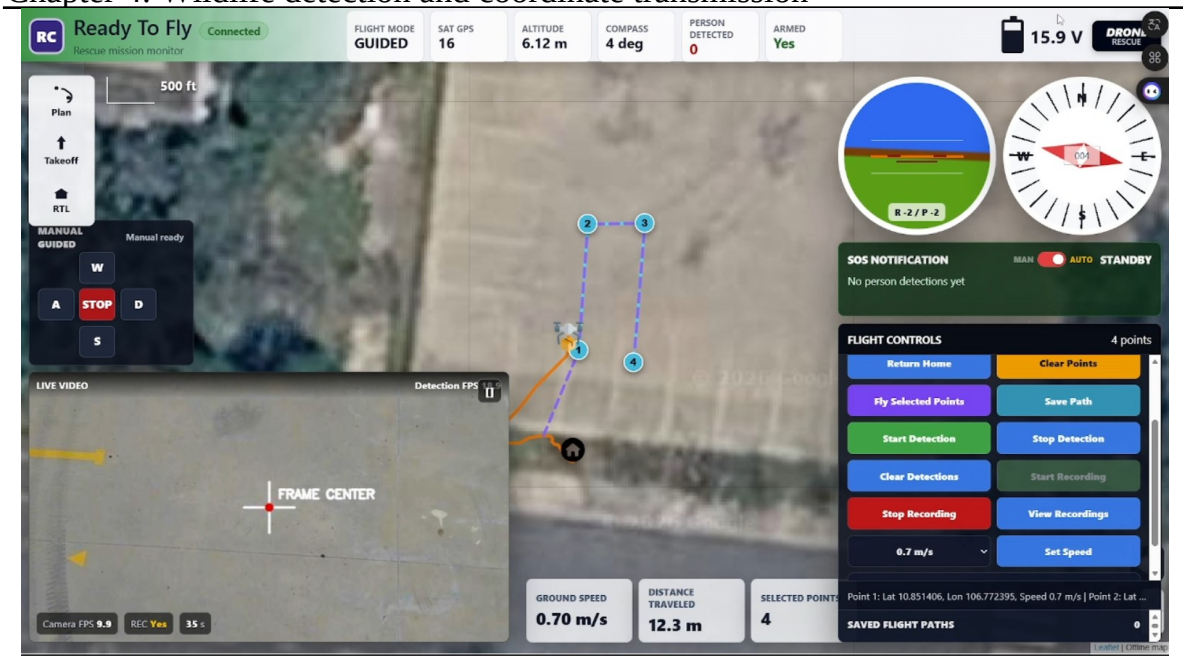
Using the UAV GPS position as the detected animal's position is a coarse approximation only when the target lies close to the camera nadir and the terrain assumptions hold. A calibrated camera model, attitude, altitude and ground intersection calculation are required for a more precise estimate, followed by comparison with surveyed target coordinates.

Input	Role in coordinate estimation
Bounding box center	Determines animal position in the image frame
Frame center	Reference point for camera alignment
UAV GPS latitude and longitude	Base position for geographic coordinate estimation
UAV altitude	Used to convert image offset into approximate ground distance
UAV heading	Used to rotate the relative offset into the geographic direction
Timestamp	Records when the animal was detected

Table 4.5. Input data used for estimating detected animal coordinates.

4.6 Sending Detection Results to the Ground Control Station

The proposed record contains label, confidence, bounding-box center, timestamp and UAV telemetry. An estimated animal coordinate should be included only when the localization calculation and its uncertainty are available.



The source draft lists several possible links (MAVLink, WebSocket and telemetry), but does not document the exact deployed channel or a measured end-to-end delay. Figure 5.6 shows a prototype ground interface; communication performance remains to be tested.

Field	Example value	Purpose
label	animal	Detected class name
confidence	Model output; example only	Reliability of detection
bbox_center	(xc, yc)	Animal position in image
uav_lat, uav_lon	Current UAV GPS	Base geographic reference
animal_lat, animal_lon	Only after geolocation validation	Coordinate sent to ground station
altitude	Current UAV altitude	Supports localization calculation
timestamp	Detection time	Data logging and mission review

Table 4.6. Example data packet transmitted from the UAV to the ground control station.

CHAPTER 5: EXPERIMENTS AND EVALUATION OF RESULTS

This chapter evaluates the two-class detector using the supplied validation figures. It reports image detection metrics and illustrates a prototype ground interface. It does not present an independent field evaluation of UAV flight, target geolocation or conservation survey performance.

5.1.1 Experimental Setup

The available experimental evidence comprises a labeled image dataset view, a confusion matrix, confidence curves, a precision-recall curve, validation predictions and an interface screenshot. Dataset size, split method, model variant, repeated-run variability and ground-truth field coordinates are not supplied.

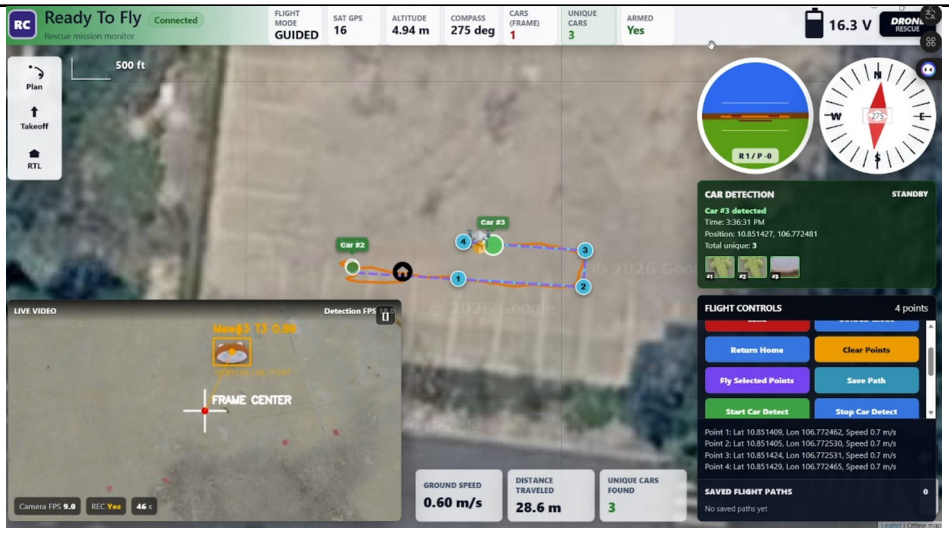
Table 5.1: Experimental setup for YOLO26 wildlife detection

Item	Description
Detection model	YOLO26 object detection model
Dataset platform	Roboflow dataset with manually annotated bounding boxes
Input source	Annotated image frames; flight provenance unverified
Detection classes	Crocodylidae and Mew
Model output	Bounding box, class name, confidence score, and target center
Evaluation outputs	Confusion matrix, F1-confidence curve, precision-confidence curve, precision-recall curve, and qualitative validation results
System output	Detected object information and candidate detection record for ground-station review

5.2 Dataset and Training Procedure

The annotated images were organized as an object-detection dataset in Roboflow. The validation figures identify two labels, Crocodylidae and Mew. These labels are preserved as recorded; they are not renamed as horses, elephants or gaur. Their provenance and the biological meaning of Mew require the original dataset documentation. The recorded metrics apply only to this detector experiment.

For the supplied validation outputs, a prediction is matched to an annotation using class identity and an intersection-over-union criterion. The displayed AP values use IoU 0.5; the exact confidence and matching settings for the confusion matrix should be retained from the validation run.



5.3 Quantitative Evaluation of the Trained Model

The supplied confusion matrix and curves describe validation performance for the two recorded labels. These results quantify detector behavior on that dataset, including missed annotations and background false positives; they do not establish generalization to an independent field site.

5.3.1. Confusion Matrix Analysis

The confusion matrix displays predicted labels on the vertical axis and true labels on the horizontal axis. The two diagonal cells count correct detections; the background row counts missed annotated targets, and the background column counts detections that do not match a target annotation.

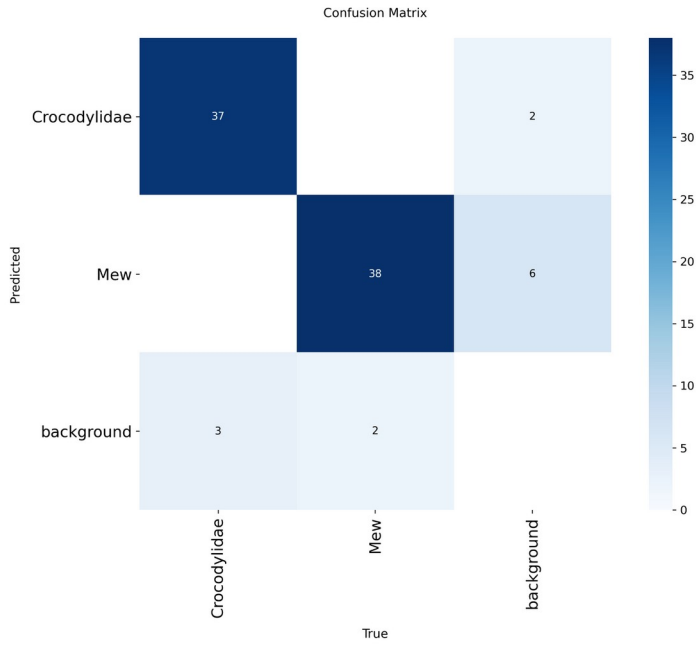


Figure 5.1. Validation confusion matrix for the two dataset labels.

The matrix shows 37 correct Crocodylidae detections and 38 correct Mew detections. It also shows three and two misses, respectively, plus two false Crocodylidae and six false Mew detections on background. The classwise and pooled precision and recall in Table 5.2 are consistent with these counts. AP@0.5 comes from the precision-recall curves, not from a single confusion matrix.

Table 5.2. Validation counts and derived precision and recall

Class	TP	FP	FN	Precision	Recall	AP@0.5
Crocodylidae	37	2	3	94.9%	92.5%	0.961
Mew	38	6	2	86.4%	95.0%	0.965
Pooled / mean AP	75	8	5	90.4%	93.8%	0.963

5.3.2 F1-Confidence Curve

The displayed F1-confidence curve peaks at approximately 0.92 at confidence 0.285. This is a threshold candidate for the supplied validation set. An operating threshold for a new habitat should be selected using its own false-positive and missed-target costs.

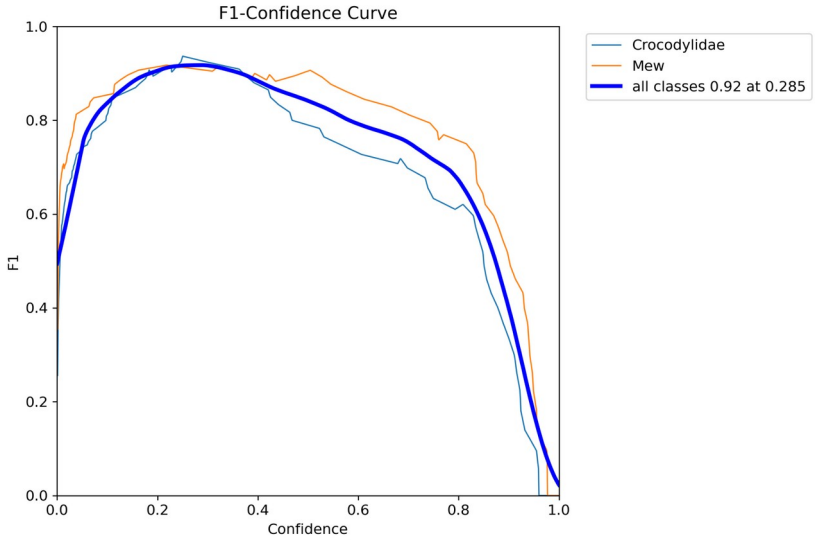


Figure 5.2. F1-confidence curves of the trained detector.

5.3.3 Precision-Confidence Curve

The displayed precision-confidence curve reaches 1.00 at approximately 0.859 for the combined classes. This is an observed validation point, not a guarantee that future high-confidence predictions will all be correct. Increasing the threshold generally reduces the number of retained detections and may also reduce recall.

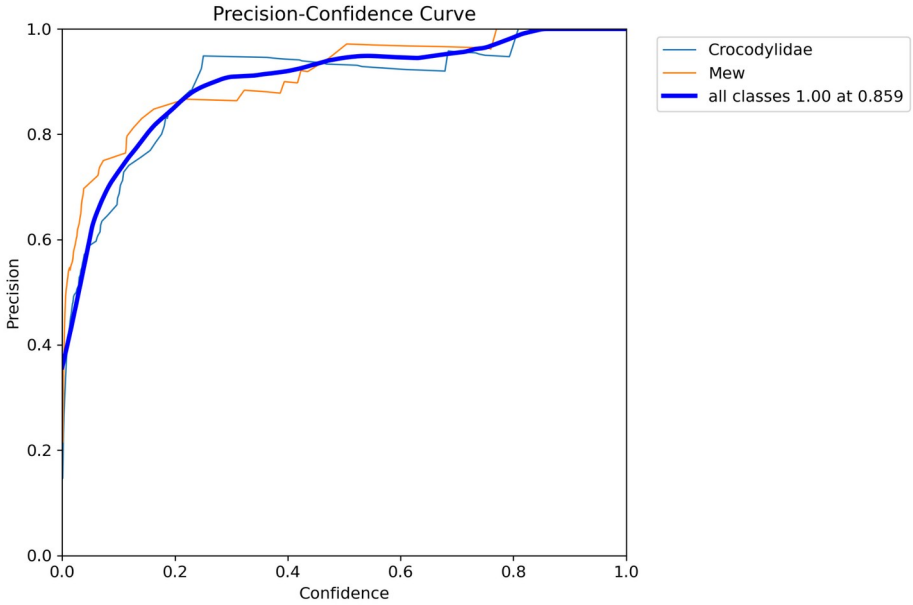


Figure 5.3: Precision-confidence curve of the trained YOLO26 model

5.3.4 Precision-Recall Curve

The precision-recall chart reports AP@0.5 of 0.961 for Crocodylidae and 0.965 for Mew; their mean is 0.963. These values summarize the displayed validation set at IoU 0.5. Additional independent images and stricter IoU thresholds are needed to judge

robustness.

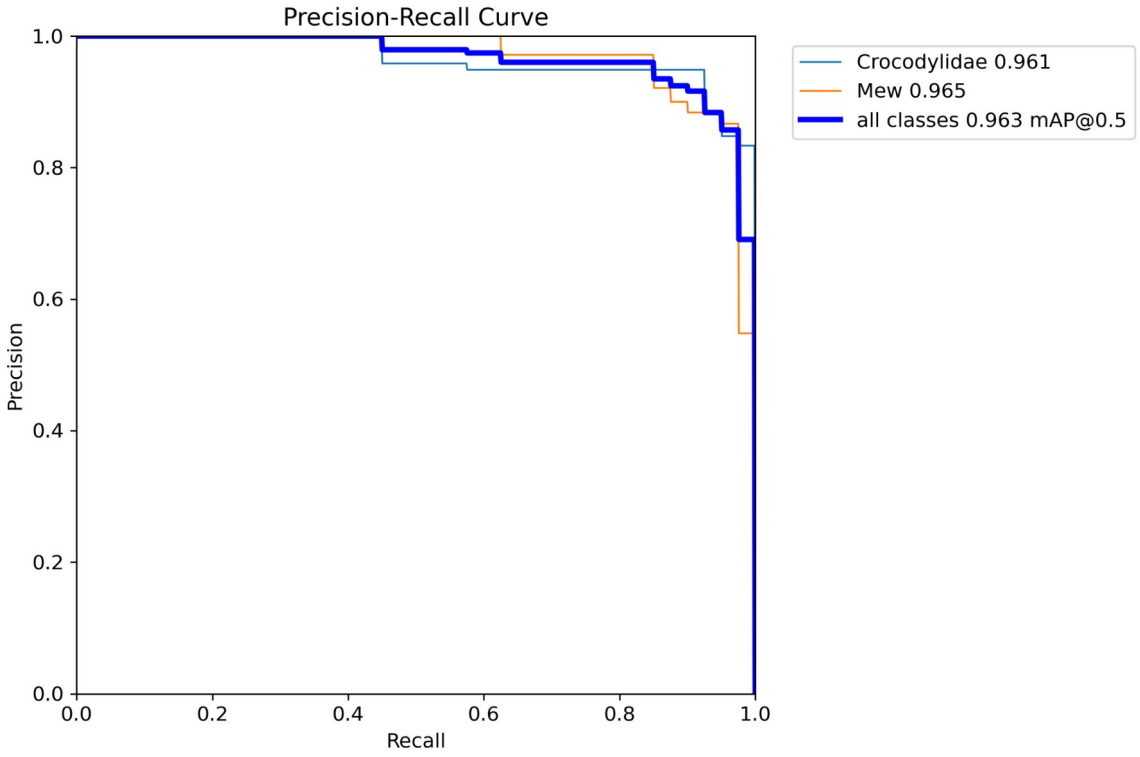


Figure 5.4: Precision-recall curve and mAP@0.5 of the YOLO26 detection model.

Table 5.3: Main evaluation indicators of the trained model

Metric	Result	Meaning
Best F1 score	0.92 at confidence 0.285	Balanced point between precision and recall
Best precision point	1.00 at confidence 0.859	High-confidence detections are very reliable
AP@0.5 - Crocodylidae	0.961	Detection quality of Crocodylidae class
AP@0.5 - Mew	0.965	Detection quality of Mew class
mAP@0.5 - all classes	0.963	Overall object detection performance at IoU 0.5

5.4 Qualitative Detection Results

The validation batch shows bounding-box predictions for both labels in repeated-looking frames. It demonstrates image-level localization in those examples. The frames' origin and independence from training are not documented, so this figure should not be treated as proof of general field performance.

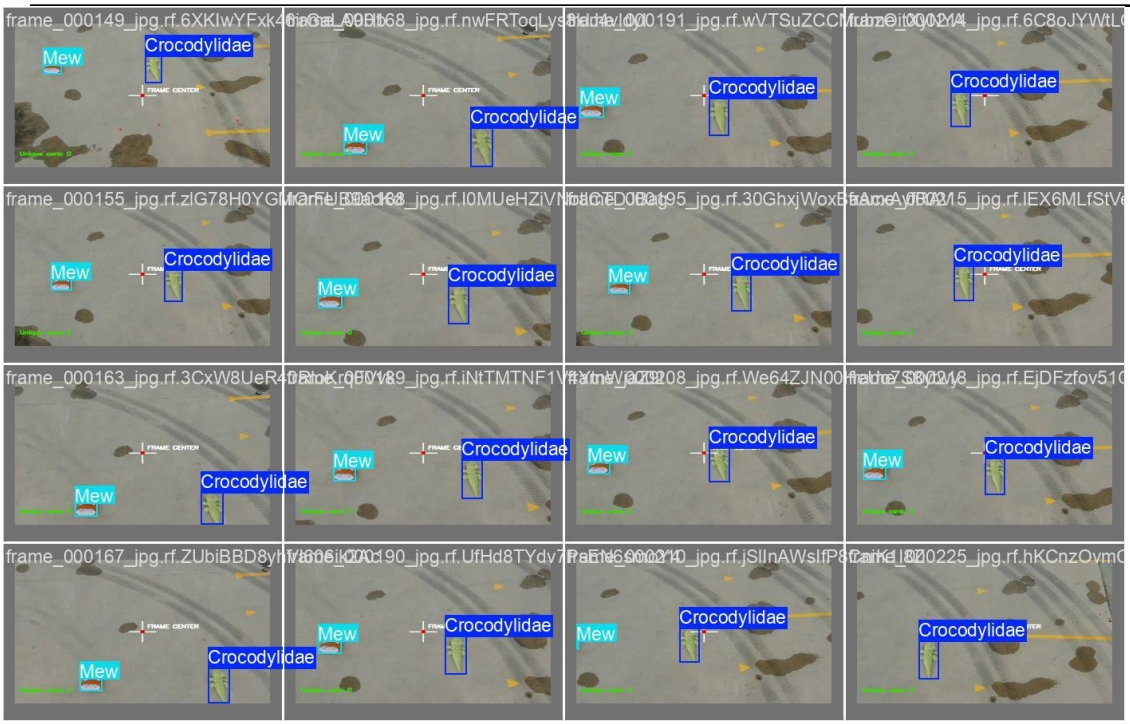


Figure 5.5. Validation batch showing YOLO26 predictions on supplied image frames.

These examples illustrate that the detector can place bounding boxes around small targets in the supplied frames. To assess performance during a real survey, future tests should include diverse habitats, camera heights, lighting, occlusion, false-positive review and independently labeled data.

5.5 UAV Monitoring Interface and Ground Control Station Integration

The screenshot demonstrates a prototype dashboard with map, flight telemetry and video panels. It does not document a synchronized flight log or a comparison of displayed animal coordinates with surveyed ground truth. The proposed workflow is to attach detection records to telemetry and let an operator review candidate observations.

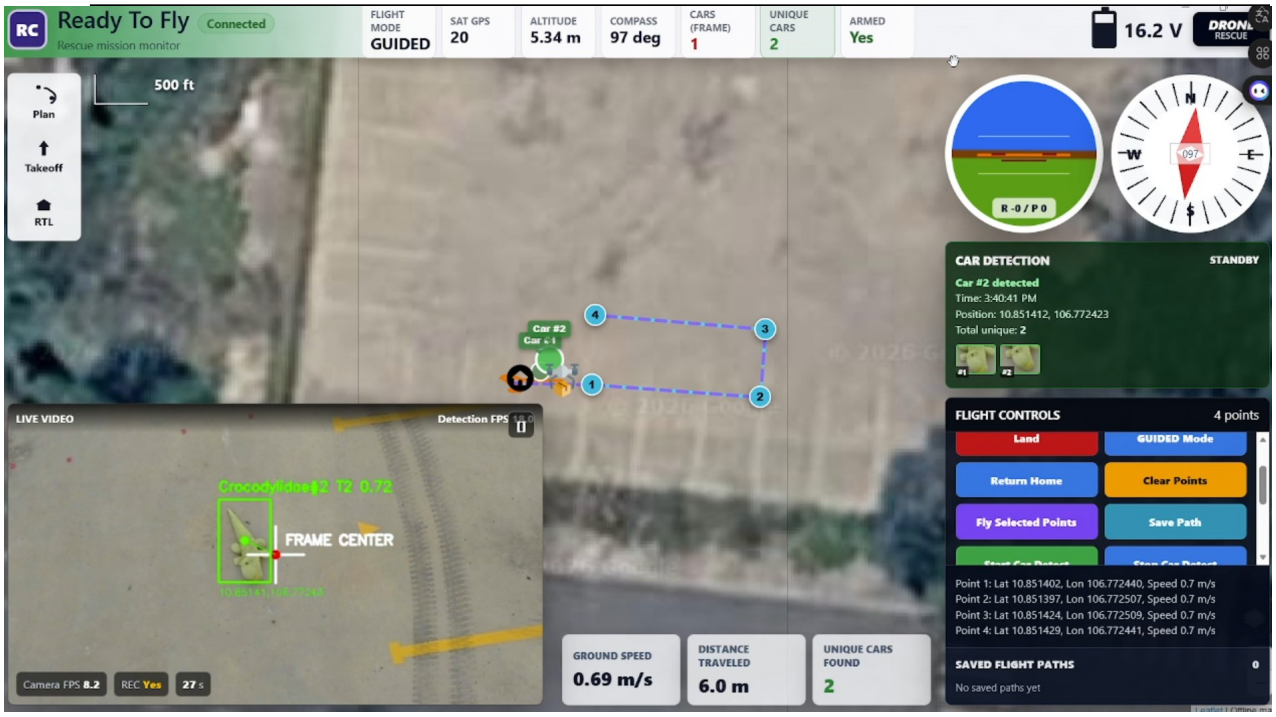


Figure 5.6. Prototype monitoring interface layout with legacy labels; this screenshot is not evidence of species detection or verified animal coordinates.

Some labels in the prototype dashboard appear to belong to an earlier application. The interface should replace those labels with the confirmed animal class names and display coordinate uncertainty or a “location unverified” state until a geolocation test is complete.

CHAPTER 6: CONCLUSION AND FUTURE DEVELOPMENT

6.1 Conclusion

The project combines a physical camera-equipped quadcopter, animal-observation field work and an image-detection workflow, motivated in part by the student's interest in horseback riding. The supplied YOLO26 validation figures record 37 correct Crocodylidae detections and 38 correct Mew detections, with mAP@0.5 of 0.963 for those two labels. These results describe that detector experiment; they do not establish recognition accuracy for horses, elephants or gaur, or operational conservation-survey performance.

The system design associates image detections with UAV telemetry and shows a prototype ground interface. The supplied materials do not establish the accuracy of estimated animal coordinates, the reliability of an onboard live pipeline or the performance of an ecological field survey.

The prototype offers a basis for further UAV-supported observation. Before conservation deployment, the team should document the dataset and class definitions, test with independent field imagery, verify camera-to-ground geometry and measure flight, communication and geolocation performance while minimizing animal disturbance.

6.2 Development direction

Next steps include confirming the second class label and dataset provenance, retaining the complete training configuration, and testing on independent images from different habitats. Report AP at multiple IoU thresholds as well as false detections per survey time and sensitivity at a selected operating threshold.

A field validation should use surveyed ground targets to measure coordinate error, record camera calibration and synchronization, and verify onboard inference rate and telemetry latency. Site permissions and species-specific disturbance protocols should guide flight height and route; thermal imaging is a possible later extension after these baseline tests.

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